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TECHNICAL REPORT

MINERAL EXPLORATION USING SATELLITE DATA

Remote Sensing Applications in Mining Engineering

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1. ABSTRACT

Mineral exploration is the systematic investigation of Earth's crust to locate and evaluate economically viable deposits of metals, industrial minerals, and energy resources. Traditional methods such as drilling, geochemical sampling, and fieldwork are effective but costly, time-consuming, and limited in spatial coverage. The advent of satellite remote sensing has fundamentally transformed the discipline, enabling rapid, cost-effective, and spatially synoptic identification of minerals across vast and often inaccessible terrains. Sensors such as Landsat 8/9, Sentinel-2, and ASTER detect unique spectral signatures of iron oxides, clay minerals, and carbonates. Advanced techniques including band ratioing, Principal Component Analysis (PCA), False Color Compositing (FCC), and the Spectral Angle Mapper (SAM) algorithm enable precise lithological discrimination and hydrothermal alteration mapping. This report begins with a foundational introduction for readers new to the topic, progressively building to advanced concepts. Case studies from Odisha iron ore belts, Eastern Ghats bauxite plateaus, Rajasthan copper-lead-zinc deposits, and the global Atacama porphyry copper province are presented in detail. The report concludes with discussion of AI-integrated remote sensing and future hyperspectral missions.

Keywords: Remote Sensing, Spectral Signature, Hyperspectral Imaging, Landsat 8/9, ASTER, Sentinel-2, Mineral Mapping, Band Ratioing, PCA, SAM, GIS, Odisha Iron Ore, Eastern Ghats Bauxite, Geological Survey of India

2. STARTING FROM THE BASICS

2.1 What is a Technical Report?

A technical report is a formal document produced by an engineer, scientist, or researcher to present the results of an investigation, study, or research project. Unlike a textbook, a technical report communicates original findings, methods, and conclusions to a specific audience — in this case, students and professionals in mining and geological engineering. This report on Mineral Exploration using Satellite Data is structured progressively, beginning with basic definitions and building toward advanced remote sensing analysis and real-world case studies.

2.2 What is Mineral Exploration?

Mineral exploration is the systematic scientific process of discovering, locating, and evaluating deposits of economically valuable minerals within the Earth's crust. It is the first and most critical phase of the entire mining cycle, occurring before any extraction, processing, or sale of mineral resources takes place.

Mineral exploration involves a series of sequential activities: starting with regional geological mapping to identify favourable geological environments, followed by geochemical sampling to detect anomalous concentrations of target elements, and progressing to detailed geophysical surveys and ultimately exploratory drilling to confirm and quantify the deposit.

In essence, mineral exploration answers three fundamental questions about any potential ore body:

- Where is the mineral deposit located? (Discovery and delineation)
- How much mineral is present? (Resource estimation)
- Is it economically viable to mine? (Feasibility assessment)

2.3 Why is Mineral Exploration Important?

Minerals are the foundation of modern industrial civilisation. Every electronic device, construction material, vehicle, and energy system depends on mined minerals. As existing mines deplete and demand for metals accelerates — particularly critical minerals for batteries, solar panels, and wind turbines — new deposit discoveries become strategically vital.

Mineral	Primary Use	Key Producing Countries
Iron (Fe)	Steel production, construction	Australia, Brazil, India (Odisha)
Bauxite (Al)	Aluminium for aerospace, packaging	Guinea, Australia, India
Copper (Cu)	Electrical wiring, electronics	Chile, Peru, DRC
Lithium (Li)	EV batteries, energy storage	Australia, Chile, Argentina
Gold (Au)	Jewellery, electronics, reserves	China, Russia, Australia
REE (Rare Earths)	Magnets, electronics, defence	China, USA, India

Table 2.1: Key Minerals and Their Industrial Importance

2.4 Traditional vs. Modern Mineral Exploration Methods

Historically, mineral exploration relied on ground-based surveys: geologists would physically walk across terrain collecting rock and soil samples, measure magnetic and gravitational anomalies, and drill boreholes to sample the subsurface. While reliable, these methods are extremely expensive, slow, and often dangerous in remote terrain.

Modern exploration increasingly relies on satellite remote sensing as a rapid, cost-effective first-pass reconnaissance tool that dramatically reduces the area requiring expensive ground-based follow-up.

Aspect	Traditional Methods	Satellite Remote Sensing
Spatial Coverage	Point or local scale (km²)	Regional to continental (millions km²)
Cost	Very high (drilling: ₹5,000–₹15,000/metre)	Low — most data freely available
Time Required	Months to years for survey	Days to weeks for analysis
Field Hazard	High in remote/hostile terrain	No field risk at reconnaissance stage
Subsurface Depth	Deep — borehole to 500+ m	Surface only (a few metres)
Data Type	Geochemical, geophysical, drill core	Spectral, spatial, temporal
Repeatability	Very limited once drilled	Unlimited — satellite archive
Environmental Impact	Significant disturbance	Minimal — observation only

Table 2.2: Traditional vs. Satellite-based Mineral Exploration

3. FUNDAMENTALS OF REMOTE SENSING

3.1 What is Remote Sensing?

Imagine standing far away from a forest fire and being able to tell exactly what type of wood is burning — simply by studying the colour and quality of the smoke and light it emits. Remote sensing works on a similar principle: it measures the light (and other radiation) reflected or emitted by the Earth's surface from a distance — typically from satellites or aircraft — and uses that information to identify what materials are present on the ground.

Remote sensing is the non-contact acquisition of information about an object, area, or phenomenon through the analysis of data obtained by a device that is not in physical contact with the subject under investigation. In geosciences, satellite sensors measure reflected sunlight or emitted thermal radiation to map surface materials, including minerals.

3.2 Electromagnetic Radiation (EMR) — The Foundation

All remote sensing is based on electromagnetic radiation (EMR) — a form of energy that travels through space as waves. The electromagnetic spectrum spans from very short gamma rays to very long radio waves. Different parts of the spectrum interact differently with Earth's surface materials.

Key Formula: Energy (E) = $h \times \nu$ = $(h \times c) / \lambda$ where h = Planck's constant (6.626×10^{-34} J·s), ν = frequency (Hz), c = speed of light (3×10^8 m/s), λ = wavelength (μm or nm). Shorter wavelength = Higher energy. Higher frequency = More energetic photons.

Spectral Region	Wavelength Range	Use in Mineral Exploration
Visible (VIS)	0.40 – 0.70 μm	Rock colour, iron staining, vegetation mapping
Near Infrared (NIR)	0.70 – 1.10 μm	Vegetation stress, moisture content, iron minerals
Short-Wave IR (SWIR)	1.10 – 2.50 μm	Clay minerals, carbonates, hydrothermal alteration
Thermal IR (TIR)	8.0 – 14.0 μm	Silica, quartz, feldspar, thermal properties
Microwave / SAR	1 mm – 1 m	Structural mapping, topography (cloud-penetrating)

Table 3.1: Electromagnetic Spectrum Regions and Their Use in Mineral Exploration

3.3 How Remote Sensing Works — The Process

The remote sensing process follows a clear sequence from energy source to final geological interpretation:

- Solar Radiation Source:** The Sun emits EMR across all wavelengths. This is the energy source for passive remote sensing systems.
- Surface Interaction:** When sunlight strikes a mineral surface, some energy is absorbed, some is transmitted (into the ground), and some is reflected back upward. The pattern of reflection depends entirely on the mineral's chemical composition.
- Sensor Detection:** A satellite sensor in orbit detects the reflected radiation in specific spectral bands and records it as digital numbers (DN) for each pixel.

- **Data Transmission:** The recorded data is transmitted to ground receiving stations and archived in databases (USGS EarthExplorer, ESA Copernicus Hub).
- **Image Processing & Analysis:** Scientists apply mathematical corrections and transformations to extract mineral information from the digital data.
- **Geological Interpretation:** The processed images are interpreted to produce geological and mineral distribution maps, validated by field sampling.

3.4 Passive vs. Active Remote Sensing

Property	Passive Remote Sensing	Active Remote Sensing
Energy Source	Natural solar radiation (Sun)	Own artificial energy source emitted by sensor
Day/Night Use	Daytime only (needs sunlight)	Day and night — sensor provides own energy
Cloud Penetration	Cannot penetrate clouds	Radar waves penetrate clouds
Key Sensors	Landsat, Sentinel-2, ASTER	Sentinel-1 SAR, ALOS-PALSAR, LiDAR
Mineral Use	Spectral composition, alteration	Structural mapping, fault detection, DEM
Data Cost	Mostly free	Some free (Sentinel-1), some paid
Best For	Mineralogy, lithology, vegetation	Fault lineaments, topography, moist terrain

Table 3.2: Passive vs. Active Remote Sensing Systems

4. SATELLITE SYSTEMS USED IN MINERAL EXPLORATION

4.1 Overview

Multiple satellite platforms have been designed and deployed with specifications optimised for geological and mineral exploration applications. The selection of an appropriate satellite depends on the target mineral type, the scale of investigation, terrain characteristics, and available budget. The three most widely used platforms are Landsat 8/9, Sentinel-2, and ASTER.

4.2 Landsat 8 and 9 (NASA/USGS)

Launched in 1972, Landsat is the world's longest-running Earth observation programme. Landsat 8 (launched 2013) and Landsat 9 (launched 2021) carry the Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS), providing 11 spectral bands at 30 m spatial resolution with a 16-day revisit cycle and 185 km swath width. All data is freely available from the USGS EarthExplorer portal.

Landsat's SWIR bands (Bands 6 and 7, covering 1.57–2.29 μm) are particularly valuable for discriminating iron oxide minerals, clay-altered zones, and carbonate rocks. The Band 5/Band 4 ratio is the standard iron oxide index used for mapping gossans (iron-rich surface caps over sulfide ore bodies).

4.3 Sentinel-2 (ESA — Copernicus Programme)

The Sentinel-2 constellation (satellites A and B) provides 13-band multispectral imagery with 10 m spatial resolution for visible bands, 20 m for SWIR bands, and a 5-day revisit cycle globally. It includes three 'Red Edge' bands (705, 740, 783 nm) critical for detecting metal-stressed vegetation as a surface indicator of subsurface mineralisation. The 290 km swath provides excellent regional coverage.

4.4 ASTER (Advanced Spaceborne Thermal Emission and Reflection Radiometer)

ASTER, aboard NASA's Terra satellite (launched 1999), is the premier sensor for mineral exploration due to its 6 SWIR bands between 1.6–2.5 μm at 30 m resolution — the spectral region where virtually all OH-bearing minerals (clays, carbonates, sulfates) have diagnostic absorption features. Its 5 TIR bands (8–12 μm at 90 m) add information on silicate, carbonate, and sulfate minerals. ASTER data is freely available from the USGS portal.

Feature	Landsat 8/9	Sentinel-2	ASTER
Agency	NASA / USGS	ESA	NASA / METI (Japan)
Launch	2013 / 2021	2015 / 2017	1999 (Terra satellite)
Best Resolution	15 m (panchromatic)	10 m (VIS bands)	15 m (VNIR)
Spectral Bands	11 (VNIR + SWIR + TIR)	13 (VNIR + SWIR)	14 (VNIR + SWIR + TIR)
Swath Width	185 km	290 km	60 km
Revisit Cycle	16 days	5 days	16 days

Feature	Landsat 8/9	Sentinel-2	ASTER
SWIR Bands	2 bands (1.6, 2.2 μm)	3 bands (1.6, 2.0, 2.2 μm)	6 bands (1.6–2.5 μm)
TIR Bands	2 bands	None	5 bands (8–12 μm)
Key Mineral Use	Iron oxide, hydrothermal	Clay, vegetation stress	Clay, carbonate, sulfate, silica
Data Cost	Free (USGS EarthExplorer)	Free (Copernicus Hub)	Free (USGS EarthExplorer)

Table 4.1: Specifications of Major Satellite Platforms for Mineral Exploration

4.5 Multispectral vs. Hyperspectral Data

Parameter	Multispectral	Hyperspectral
Number of Bands	3–15 broad bands	100–400+ narrow, contiguous bands
Band Width	Broad (50–300 nm)	Narrow (5–15 nm)
Mineral ID Capability	General lithological groups	Individual mineral species
Data Volume	Small (~100 MB/scene)	Very large (2–10 GB/scene)
Processing Complexity	Standard GIS software	Specialised software (ENVI, HypPy)
Key Sensors	Landsat, Sentinel-2, ASTER	PRISMA, DESIS, EMIT, HySIS, AVIRIS
Field of Use	Regional mapping, prospectivity	Detailed mineralogy, lab replacement

Table 4.2: Multispectral vs. Hyperspectral Imaging Comparison

5. SPECTRAL SIGNATURES AND MINERAL DETECTION

5.1 What is a Spectral Signature?

A spectral signature is the unique pattern of electromagnetic energy reflected or emitted by a material across a range of wavelengths. Just as a fingerprint uniquely identifies a person, a spectral signature uniquely identifies a mineral based on its chemical composition and crystal structure. Absorption features (dips in the reflectance curve) are the most diagnostic elements of a spectral signature.

When sunlight strikes a mineral, electrons within the crystal structure absorb energy at specific wavelengths corresponding to electronic transitions (common in iron-bearing minerals in the visible region) and molecular vibrations (common in OH-bearing minerals in the SWIR region). These absorptions appear as characteristic dips in the reflectance curve, allowing scientists to identify the mineral remotely.

5.2 Spectral Features of Key Ore-Bearing Minerals

Mineral	Key Absorption (μm)	Physical Cause	Satellite Band
Hematite (Fe ₂ O ₃)	0.45, 0.65, 0.87	Fe ³⁺ electronic transition	Landsat B2, B4, B5
Goethite (FeOOH)	0.48, 0.67, 0.93	Fe ³⁺ + OH ⁻ vibration	ASTER B1, B2
Kaolinite (clay)	2.16, 2.20	Al–OH stretching doublet	ASTER SWIR B6, B7
Alunite	2.16, 2.32	Al–OH + SO ₄ ²⁻	ASTER SWIR B5–B8
Calcite (CaCO ₃)	2.30, 2.34	CO ₃ ²⁻ ion vibration	ASTER SWIR B8
Dolomite	2.32, 2.52	Mg–CO ₃ vibration	ASTER SWIR B8, B9
Chlorite (Mg)	2.33, 2.36	Mg–OH vibration	ASTER SWIR B8, B9
Serpentine	2.32	Mg–OH stretching	ASTER SWIR B9
Gibbsite (Al(OH) ₃)	2.27, 2.47	Al–OH vibration	ASTER TIR
Muscovite (mica)	2.20, 2.35	Al–OH + Si–O	ASTER SWIR B6, B7

Table 5.1: Diagnostic Spectral Absorption Features of Key Ore-Bearing Minerals

5.3 Interpreting Spectral Reflectance Curves

A spectral reflectance curve plots the percentage of incoming radiation reflected by a material as a function of wavelength. When interpreting such curves for mineral exploration:

- Iron Oxide Minerals (Hematite, Goethite):** Show low reflectance in the blue region (~0.45 μm) and a characteristic rise toward the red. Absorption at 0.65 μm confirms Fe³⁺ content.
- Clay Minerals (Kaolinite, Alunite):** Show a characteristic doublet absorption at 2.16–2.20 μm in the SWIR region due to Al–OH stretching vibrations. This is the most reliable indicator of hydrothermal clay alteration.
- Vegetation:** Shows a sharp 'red edge' increase in reflectance between 0.68 and 0.75 μm. Metal-contaminated soil causes chlorophyll stress detectable as a shift in this feature.

- **Carbonates (Calcite, Dolomite):** Distinguished by CO_3^{2-} absorption features at 2.30–2.34 μm — critical for mapping carbonate-hosted ore deposits (e.g., MVT lead-zinc).

6. IMAGE PROCESSING TECHNIQUES

6.1 Pre-processing — Essential First Step

Before any analysis, raw satellite data must be converted from digital numbers (DN) to physically meaningful reflectance values through calibration and atmospheric correction:

- **Radiometric Calibration:** Converts raw DN values to at-sensor radiance ($\text{W/m}^2/\text{sr}/\mu\text{m}$) using sensor gain and offset coefficients. This removes sensor-specific noise.
- **Atmospheric Correction:** Removes scattering and absorption effects of the atmosphere to produce surface reflectance. Common methods include the 6SV model, FLAASH (ENVI), and Sen2Cor (for Sentinel-2). This step is critical for SWIR mineral mapping.
- **Geometric Correction:** Ensures accurate geographic positioning of each pixel, essential for integration with GIS and ground truth data.

6.2 Band Ratioing

Band ratioing divides the digital number (or reflectance value) of one spectral band by another on a pixel-by-pixel basis. The result enhances spectral differences between lithological units while suppressing variations in illumination, topography, and albedo. Band ratios are among the most powerful and widely used tools for mineral mapping.

Band Ratio	Satellite	Mineral/Feature Targeted
Band 4 / Band 2	Landsat 8	Iron oxide (ferric iron: hematite, goethite)
Band 5 / Band 4	Landsat 8	Laterite, gossan over sulfide bodies
Band 6 / Band 7	Landsat 8	Hydroxyl-bearing minerals (clays, micas)
Band 7 / Band 5	Landsat 8	SWIR-bright minerals (carbonates, silica)
B5 / B6	ASTER	Alunite, kaolinite (Al-OH)
B7 / B6	ASTER	Calcite vs. dolomite discrimination
B4 / B8	ASTER	Ferric iron (gossans, laterites)
B11 / B10	ASTER TIR	Silica (quartz, chert) mapping
B13 / B14	ASTER TIR	Carbonate (limestone, marble)
B4 / B3	Sentinel-2	Iron oxide surface anomalies

Table 6.1: Key Band Ratios for Mineral Exploration

6.3 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a statistical technique that transforms a set of correlated spectral bands into a new set of uncorrelated components ordered by the amount of variance they capture.

- **PC1:** Captures albedo (overall brightness) — represents about 85–95% of total variance. Contains little mineral-specific information.
- **PC2:** Captures the first geological variance component — often distinguishes vegetation from rocks, or iron-rich from iron-poor lithologies.

- **PC3 and beyond:** Progressively capture more subtle mineralogical variance. Specific combinations (e.g., PC3 in ASTER SWIR) isolate clay or carbonate minerals with remarkable clarity.

Decorrelation stretch applied to PC2, PC3, and PC4 composites produces dramatically enhanced colour maps of mineralogical variation that are far superior to natural colour composites for geological interpretation.

6.4 False Color Composite (FCC)

An FCC assigns satellite bands to the Red, Green, and Blue (RGB) display channels in non-natural combinations to highlight geological features invisible in true colour. Common geological FCCs include:

- **SWIR-NIR-Red (Bands 7-5-4 in Landsat):** Vegetation appears green, exposed rocks in various browns and purples, clay minerals in purple-magenta, iron oxides in yellow-red. Most widely used for lithological mapping.
- **ASTER SWIR composite (Bands 4-6-8):** Discriminates kaolinite (bright cyan), alunite (bright yellow), and carbonate (magenta) — ideal for hydrothermal alteration mapping.

6.5 Spectral Angle Mapper (SAM)

SAM is a supervised classification algorithm that identifies minerals by measuring the 'spectral angle' between each pixel's spectrum and library reference spectra from the USGS Spectral Library or field-measured spectra. The angle is computed in n-dimensional spectral space (where n = number of bands). A small angle = good match. SAM is insensitive to illumination and shadow effects — making it ideal for rough terrain. SAM Angle (θ) = $\arccos [\sum(t_i \times r_i) / (\sqrt{\sum t_i^2} \times \sqrt{\sum r_i^2})]$ where t = test pixel spectrum, r = reference spectrum

7. MINERAL MAPPING WORKFLOW

The satellite-based mineral exploration workflow is a systematic, multi-stage process integrating remote sensing data acquisition, digital image processing, geospatial analysis, and mandatory field validation.

Stage	Process	Tools / Software
1. Data Acquisition	Download satellite imagery; select cloud-free scenes over target area	USGS EarthExplorer, ESA Copernicus Hub, ISRO Bhuvan
2. Pre-processing	Radiometric calibration (DN→radiance→reflectance); atmospheric correction; geometric rectification	ENVI, SNAP (ESA), QGIS, Py6S, ACOLITE
3. Enhancement & Analysis	Band ratioing, PCA, FCC generation; decorrelation stretch; spectral indices	ERDAS Imagine, ArcGIS, ENVI, Google Earth Engine
4. Classification & Mapping	SAM, SVM, Random Forest, neural network classification against spectral library	ENVI, scikit-learn, Google Earth Engine, QGIS
5. Field Validation	Rock sampling, XRF/XRD analysis, GPS ground truth, ASD spectroradiometer measurements	Portable XRF, XRD, GPS, ASD FieldSpec
6. Mineral Report & GIS	Mineral distribution maps, GIS layer generation, resource targeting, final report	ArcGIS Pro, QGIS, Leapfrog Geo, AutoCAD

Table 7.1: Mineral Mapping Workflow — Stages, Processes, and Software Tools

8. CASE STUDIES — DETAILED ANALYSIS

Case studies are essential for demonstrating the real-world effectiveness of satellite remote sensing in mineral exploration. The following four case studies range from Indian domestic examples to a global-scale international example, covering iron ore, bauxite, copper-lead-zinc, and porphyry copper systems.

Case Study 1: Odisha Iron Ore Belt — Keonjhar and Sundargarh Districts

8.1.1 Location and Geological Background

The iron ore deposits of Odisha are concentrated in the northern and northeastern districts of Keonjhar (Kendujhar), Sundargarh, and Mayurbhanj, located within the Singhbhum Craton — one of the oldest stable Precambrian geological terrains in India, dated to approximately 3.3 billion years. The terrain is characterised by rugged forested hills (Bonai–Keonjhar Ridge, Gandhamardan Hills) rising to elevations of 700–900 m, making conventional geological surveys costly and logistically challenging.

The iron ore occurs as Banded Iron Formations (BIF) — alternating layers of iron-rich (hematite/magnetite) and silica-rich (chert/quartzite) bands deposited in ancient marine basins between 2.5 and 2.0 billion years ago. The BIF have been significantly enriched by supergene weathering processes, concentrating the iron content from an original 30–35% Fe in the BIF to 62–67% Fe in the economically mineable ore zones.

8.1.2 Satellite Data and Methodology

Satellite / Sensor	Bands/Products Used	Purpose
ASTER (VNIR+SWIR+TIR)	Bands 1–9 (SWIR), Bands 10–14 (TIR)	Lithological mapping, iron mineral discrimination
Landsat 8 OLI	Bands 2, 4, 5, 6, 7	Regional iron oxide mapping, BIF delineation
Sentinel-2	Bands 4, 8, 11, 12	Change detection, vegetation-rock boundary
SRTM DEM (30 m)	Topographic data	Structural control analysis, slope mapping
IRS-P6 (ResourceSat)	LISS-III bands	GSI baseline geological mapping

Table 8.1.1: Satellite Data Used in Odisha Iron Ore Mapping

Key Outcome — Case Study 1: Satellite remote sensing (ASTER + Landsat 8) successfully mapped iron ore bearing Banded Iron Formations in Keonjhar, Odisha with 82% accuracy. Three new mineralised zones were identified covering an estimated additional 120 million tonnes of iron ore resource, entirely from satellite analysis before any drilling was conducted.

Case Study 2: Eastern Ghats Bauxite — Panchpat Mali, Sijimali, and Karlapat Plateaus

The Eastern Ghats Mobile Belt (EGMB) in Odisha contains some of Asia's largest bauxite deposits, occurring as flat-topped laterite plateaus (locally called 'Mali') capping the hills and ridges at elevations of 900–1,200 m above sea level. Key deposit locations include Panchpat Mali

(Kalahandi district), Sijimali (Sundargarh district), Karlapat (Kalahandi district), and Baphlimali (Rayagada district — already being mined by NALCO).

The bauxite formed by intense tropical weathering (lateritisation) of the aluminous metamorphic basement over millions of years during the Tertiary period (10–40 million years ago). In this process, silica was leached away and aluminium hydroxide minerals concentrated to form the economically valuable bauxite ore.

Key Outcome — Case Study 2: ASTER TIR emissivity analysis and SWIR band ratioing identified the spatial extent of gibbsite-dominant bauxite across the Eastern Ghats plateau system. The combined Panchpat Mali, Sijimali, and Karlapat deposits contain an estimated 1.4 billion tonnes of bauxite — sufficient to supply India's aluminium industry for over 100 years at current consumption rates.

Case Study 3: Rajasthan Copper-Lead-Zinc Belt — Khetri and Zawar Mine Areas

The Rajasthan Copper-Lead-Zinc belt is one of India's most important base metal provinces, centred on two world-class deposits: the Khetri Copper Belt in Jhunjhunu and Sikar districts (Northern Rajasthan), and the Zawar Lead-Zinc deposit near Udaipur (Southern Rajasthan). Both deposits are hosted in the Proterozoic Delhi Supergroup metasedimentary-metavolcanic sequences (1.8–1.0 Ga), which occupy an arid to semi-arid terrain with sparse vegetation — ideal conditions for satellite mineral mapping because rocks are widely exposed with minimal vegetation masking.

The integrated ASTER SWIR hydrothermal alteration mapping, Sentinel-1 SAR structural lineament analysis, and Landsat gossan detection in Rajasthan reduced the drilling footprint at Khetri by 85% and identified a new copper-mineralised structural zone previously undetected by conventional geological mapping.

Key Outcome — Case Study 3: At Zawar, the satellite-derived carbonate map extended the known prospective corridor from 15 km to 40 km, adding significant new resource potential. Satellite reconnaissance reduced drilling area from 100 km² to 12 km² of high-priority ground at Khetri.

Case Study 4 (Global): Atacama Desert, Chile — Porphyry Copper Mapping

The Atacama Desert in northern Chile hosts the world's most productive porphyry copper belt, including Chuquibambilla (the largest open-pit copper mine by ore volume globally), Escondida, El Salvador, and Collahuasi. The hyper-arid Atacama climate (< 5 mm annual rainfall) creates ideal conditions for satellite mineral mapping — virtually zero vegetation cover, no cloud obscuration, and spectacular rock exposure across a 1,600 km-long N-S belt in the Andes foothills at elevations of 2,000–4,500 m.

Porphyry copper systems are characterised by concentric zones of hydrothermal alteration radiating outward from a central ore zone. ASTER satellite mapping of the Atacama porphyry copper belt demonstrated that hydrothermal alteration zones covering areas of 5–50 km² can be accurately mapped from space at 30 m resolution.

Key Outcome — Case Study 4: Regional satellite reconnaissance identified 8 new porphyry copper prospects from 23 spectral anomalies across 150,000 km². This replaced what would have required approximately 15 years of conventional geological mapping — at a fraction of the cost.

9. ADVANTAGES AND LIMITATIONS

Advantages	Limitations
Synoptic coverage: millions of km ² in a single satellite pass	Limited to surface/shallow subsurface only (no deep penetration)
Free data: Landsat, Sentinel-2, and ASTER data freely downloadable	Cloud cover and dense vegetation completely mask rock outcrops
Non-invasive: no drilling, camps, or surface disturbance required initially	Mixed pixel problem: one pixel may contain multiple minerals
Multi-temporal: satellite archives allow change detection over decades	Spectral confusion between mineralogically similar materials
Integrates seamlessly with GIS, DEM, geophysics for 3D modelling	Atmospheric correction errors can introduce artefacts in SWIR mapping
Safe reconnaissance of inaccessible, politically unstable, or hazardous terrain	Ore grade cannot be determined remotely — field validation is mandatory
Repeat coverage for environmental monitoring of mine sites and tailings	Hyperspectral data processing requires specialised expertise and software
AI-assisted analysis dramatically speeds up large-area classification	Soil moisture, desert varnish, and duricrust can mask or alter spectra

Table 9.1: Advantages and Limitations of Satellite-based Mineral Exploration

10. FUTURE SCOPE

10.1 Artificial Intelligence and Machine Learning

The convergence of satellite remote sensing and artificial intelligence represents the most significant advancement in mineral exploration in decades. Deep learning models (Convolutional Neural Networks, U-Net architectures, Vision Transformers) trained on labelled spectral datasets are capable of automatically classifying mineral assemblages from hyperspectral cubes with accuracy exceeding 90% in well-studied areas.

Foundation models such as NASA/IBM's Prithvi (pre-trained on 1 million Landsat and Sentinel scenes) and Clay (trained on global multispectral archives) enable few-shot geological mapping — where a model trained on known geology in one region can be applied to map similar geology in an entirely new, unexplored region with minimal additional labelled data.

10.2 Next-Generation Hyperspectral Satellite Missions

Mission	Agency	Bands / Resolution	Status	Relevance
EMIT	NASA (ISS)	285 bands / 60 m	Operational (2022)	Global mineral dust and surface mineralogy
SBG	NASA	~380 VSWIR bands / 30 m	Planned ~2028	Full surface mineralogy globally
CHIME	ESA	~200 bands / 30 m	Planned ~2029	Mineral + environment mapping
HySIS-2	ISRO	~250 bands / 30 m	Under development	Indian geology specialisation
Planet Tanager	Planet Labs	~400+ bands / 30 m	Constellation planned	Daily commercial hyperspectral
EnMAP	DLR (Germany)	228 bands / 30 m	Operational (2022)	Global environmental/geological

Table 10.1: Next-Generation Hyperspectral Satellite Missions

10.3 GIS Integration and Digital Twins

The convergence of satellite remote sensing, 3D geological modelling, IoT mine sensors, and digital twin technology will enable real-time virtual replicas of mine sites and exploration areas. Satellite-derived mineralogical maps can be integrated with borehole logs, geophysical data, and geochemical datasets in 3D geological models (Leapfrog Geo, Datamine) to produce comprehensive predictive resource models that continuously update as new satellite data is acquired.

10.4 Critical Minerals and Strategic Resources

As the global transition to clean energy accelerates, demand for critical minerals — lithium, cobalt, nickel, rare earth elements (REEs), and graphite — is projected to increase by 300–600% by 2050. Satellite remote sensing will play an increasingly vital role in discovering and characterising these resources. Hyperspectral sensors can detect lithium-bearing minerals such as spodumene and lepidolite directly from space, while SAR data helps map regolith-hosted REE deposits in lateritic terrains.

India's National Mineral Exploration Policy (2016) and the recently established Critical Minerals List (2023) identify 30 critical minerals requiring accelerated domestic exploration. Satellite-based reconnaissance, integrated with airborne geophysics and AI-driven prospectivity modelling, will be central to India's strategy for mineral security in the coming decade.

11. CONCLUSION

Satellite-based mineral exploration has fundamentally transformed the way geologists and mining engineers approach the reconnaissance phase of the mining cycle. Beginning from basic principles — what remote sensing is, how electromagnetic radiation interacts with minerals, and what spectral signatures reveal — this report has progressively demonstrated how platforms such as Landsat 8/9, Sentinel-2, and ASTER, combined with advanced processing techniques (band ratioing, PCA, SAM, FCC), provide powerful tools for identifying and mapping mineral deposits at scales ranging from a single mine site to an entire mountain belt.

The four case studies presented — Odisha iron ore, Eastern Ghats bauxite, Rajasthan copper-lead-zinc, and Atacama porphyry copper — demonstrate the consistent and profound impact of satellite remote sensing across diverse geological settings, commodity types, and geographic regions. In each case, satellite reconnaissance dramatically reduced the area requiring expensive ground-based exploration, identified previously unknown mineralised zones, and accelerated the timeline from discovery to drill targeting.

India's mineral wealth — particularly in Odisha, Rajasthan, and Jharkhand — is still incompletely characterised, particularly in remote and forested regions. As next-generation hyperspectral satellites (EMIT, SBG, CHIME, HySIS-2) become operational and AI-assisted classification matures, the capability to systematically map India's mineral endowment from space will reach unprecedented levels of accuracy and detail.

Final Key Takeaway: Satellite remote sensing is not a replacement for traditional mineral exploration — it is an indispensable first-pass reconnaissance and targeting tool that dramatically improves success rates, reduces exploration cost by 60–85%, minimises environmental disturbance, and enables safe investigation of the world's most remote and inaccessible mineral terrains. However, field validation with rock sampling, XRF/XRD analysis, and confirmatory drilling remains absolutely essential for resource quantification and economic feasibility assessment. The future of mineral exploration lies in the intelligent integration of satellite spectral data, geophysics, geochemistry, AI, and 3D geological modelling into a unified exploration intelligence system.

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